

18.3.2 Differential Circuits

In order to precisely slice input data, as seen in Fig. 18.12, a reference voltage may be transmitted, on a different signal path, along with the data. Alternatively, the data may be transmitted differentially (an input and its complement). In either case a differential amplifier is needed, Fig. 18.17. A differential amplifier input buffer (which we'll simply call an input buffer from this point on) amplifies the difference between the two inputs. In the simplest case one input to the input buffer is a DC voltage, say 0.5 V (V_{inm} in Fig. 18.17). When the other input (V_{inp}) goes above 0.5 V, the output of the buffer changes states (goes from a low to a high) or

$$V_{inp} > V_{inm} \rightarrow \text{out} = "1"$$
(18.22)

and

$$V_{inp} < V_{inm} \rightarrow \text{out} = "0"$$
(18.23)

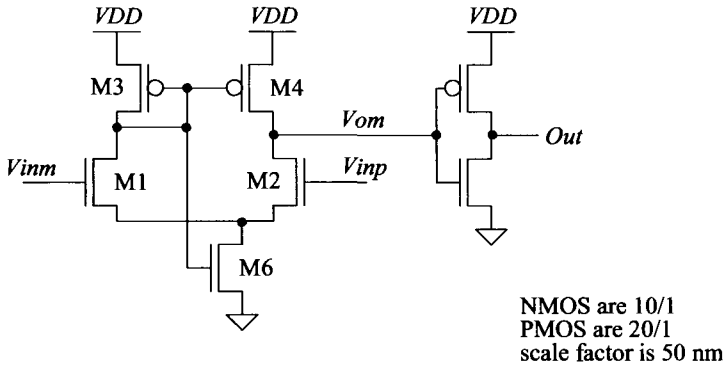


Figure 18.17 An (n-flavor) input buffer for high-speed digital design.

The diff-amp in Fig. 18.17 is based on the topologies seen in Ch. 22. This circuit is *self-biased* because no external references are used to set the current in the circuit (the gate of M6 is tied up to the gate of M3). When V_{inp} is larger than V_{inm} , the current in M2 is larger than the current in M1 ($V_{GS2} > V_{GS1}$). The current in M1 flows through M3 and is mirrored by M4 (and so M4's current is less than M2's current). This causes the diff-amp's output, V_{om} , to go towards ground (until the current in M2 equals the current in M4) and the output of the inverter, Out , to go high. Note that the gain from the V_{inp} input to the output of the circuit is larger than the gain from V_{inm} to the output (M3, being diode-connected, is a lower resistance than M4). It is generally a good idea to connect the reference voltage to the V_{inm} input.

The DC simulation results for the buffer in Fig. 18.17 are shown in Fig. 18.18. The x-axis is the V_{inp} input swept from 0 to 1 V. The V_{inm} input is held from 0 to 1 V in 200 mV steps. When V_{inm} is held at 400 mV and V_{inp} goes above 400 mV, the output changes from 0 to 1 (although there is a small offset which necessitates that V_{inp} go to approximately 415 mV before the output transitions).

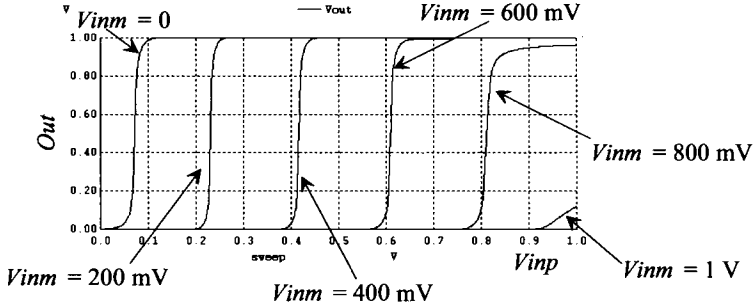


Figure 18.18 Simulating the DC behavior of the buffer in Fig. 18.17.

Transient Response

An example transient response for the buffer in Fig. 18.17 is seen in Fig. 18.19. A very small increase in V_{inp} above V_{inm} is required to make the output of the buffer switch states. We have several practical questions that should be answered with variations of this simulation. For example, what are the minimum and maximum values of V_{inm} allowed. Next, why are the delays between V_{inp} going high and going low different? Is it because of the offset seen in Fig. 18.18 (the output doesn't precisely switch at a V_{inm} 400 mV)? Let's provide some discussion concerning these concerns.

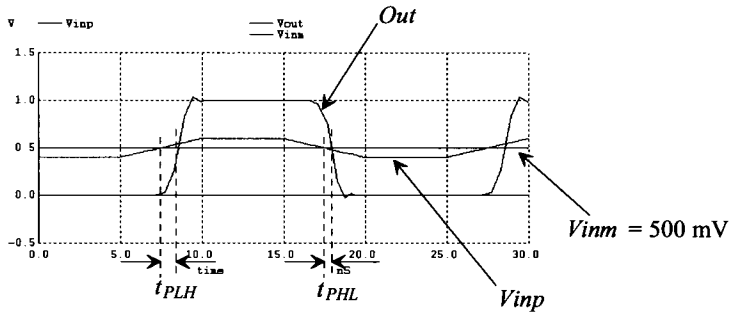


Figure 18.19 Transient response of the input buffer in Fig. 18.17 driving a 50 fF load.

Looking at Fig. 18.17, we can see that if the inputs fall below V_{THN} ($= 250$ mV here), then the circuit won't work very quickly (the MOSFETs move into the subthreshold region). So we would expect the propagation delays to increase. Indeed, as seen in Fig. 18.20, the delays go up. Ideally, the delay of the buffer is independent of power supply voltage, temperature, or input signal amplitudes (or pulse shape). To get better performance for lower input level signals, we might use the PMOS version of the buffer in Fig. 18.17, as seen in Fig. 18.21. Resimulating this buffer with the signals seen in Fig. 18.20 gives the results seen in Fig. 18.22. The delays are considerably better, however, there is an offset that appears rather large (because the output changes at the same time as V_{inp} going past V_{inm} , indicating that an offset is present). To avoid this

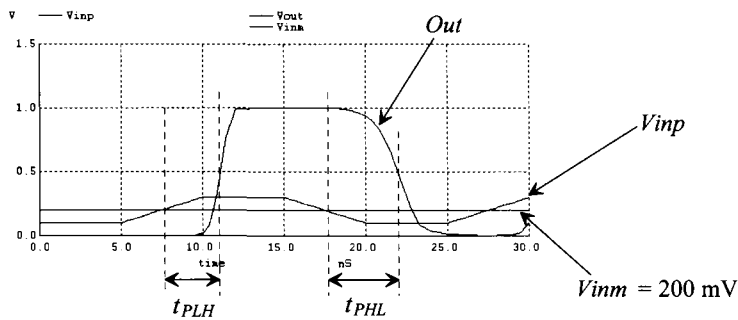


Figure 18.20 Resimulating with lower input signal voltages.

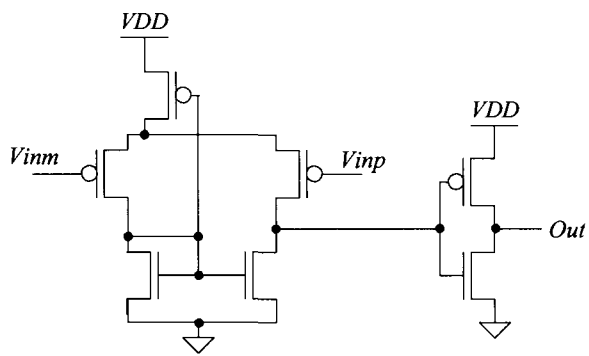


Figure 18.21 A PMOS input buffer for high-speed digital design.

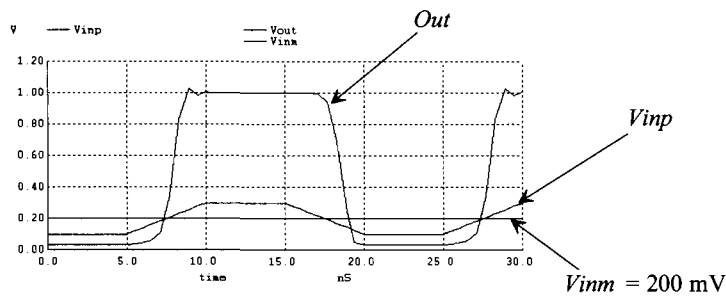


Figure 18.22 Repeating the simulation in Fig. 18.20 with the buffer in Fig. 18.21.

offset, we might use the NMOS buffer in Fig. 18.17 with the PMOS buffer in Fig. 18.21 to form a buffer that operates well with input signals approaching ground or V_{DD} . The result is seen in Fig. 18.23. By using the buffers in parallel, the complementary nature results in a buffer that is robust and works over a wide range of operating voltages. Figure 18.24 shows a DC sweep simulation for the buffer with the V_{inp} input swept and the V_{inm} changed in increments of 100 mV. Note the smaller offset.

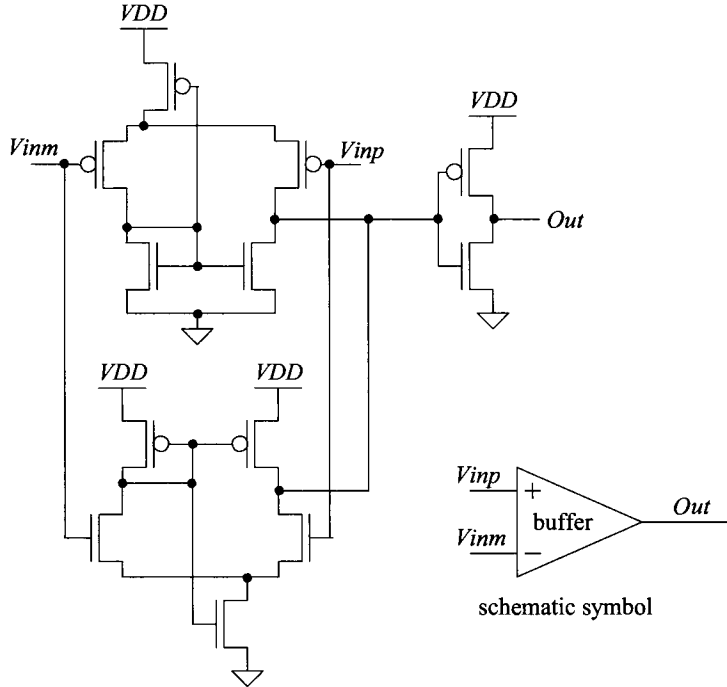


Figure 18.23 A rail-to-rail input buffer based on the topologies in Figs. 18.17 and 18.21.

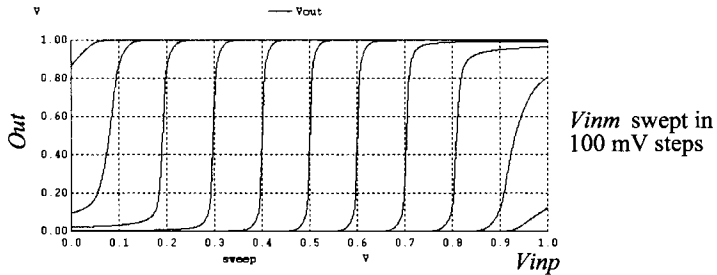


Figure 18.24 DC characteristics of the buffer in Fig. 18.23.

Example 18.2

Design a high-speed input buffer based on the topology in Fig. 18.17 with a small amount of hysteresis.

In any circuit that has hysteresis we have some positive feedback. We can introduce positive feedback into the buffer in Fig. 18.17 by adding a long L MOSFET across the output inverter, Fig. 18.25. When the output is low, this turns the long L MOSFET on and make the self-biased diff-amp work harder to pull the input of the inverter to ground. When the output is high, the long L MOSFET is off and doesn't affect the circuit. ■

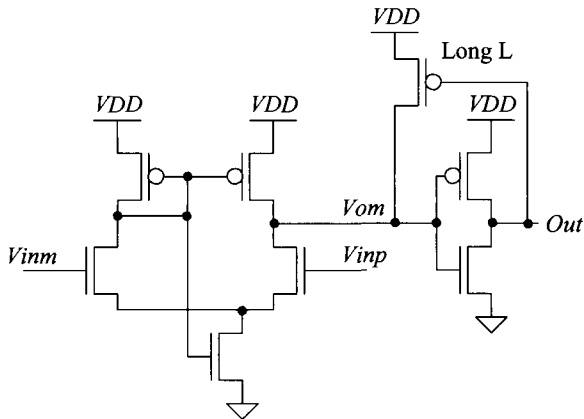


Figure 18.25 An input buffer for high-speed digital design with hysteresis.

18.3.3 DC Reference

In the previous section we assumed we had a DC voltage, V_{inm} , at precisely the correct voltage to slice the data (in the middle). In a real system the data varies and the communication channel is bandlimited (behaves like a lowpass filter). The result is that the amplitudes of the data vary depending on the data and the channel frequency response. Consider the simple RC lowpass filter and data used to model a transmission system as seen in Fig. 18.26. The ideal point where we slice the output data changes with the input data. What we need is a circuit that determines the maximum and minimum of an input waveform and outputs the average of the two to slice the buffer's input data in the center.

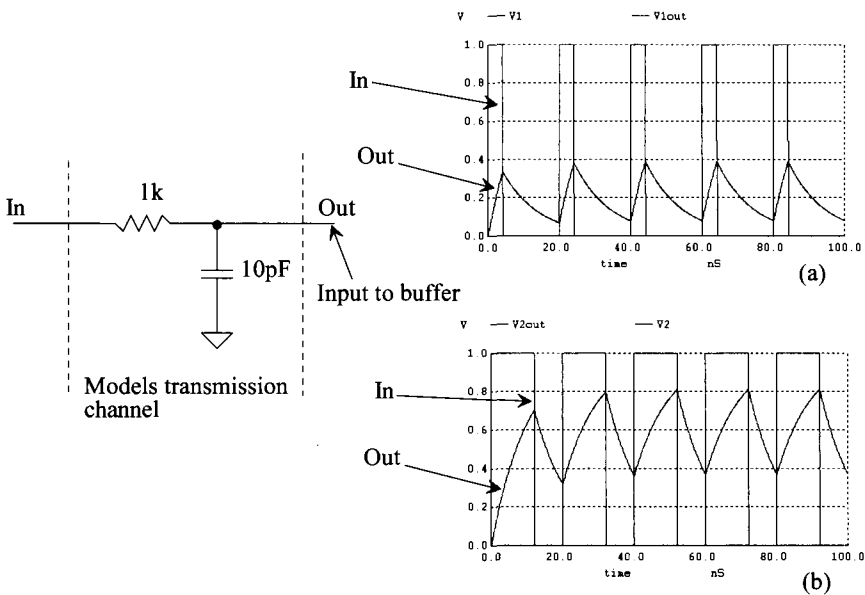


Figure 18.26 How the shape of the data changes through a communication channel.

Towards this goal consider the peak detector circuit seen in Fig. 18.27. The buffer from Fig. 18.23 can be used in this circuit. When the input, v_{in} , goes above the voltage stored on the capacitor, v_{peak} , the output of the buffer goes low, turning on the long-length MOSFET and pulling the output towards VDD . As v_{peak} approaches v_{in} , the MOSFET starts to shut off. As a result the output voltage across the capacitor, v_{peak} , corresponds to the peak voltage of the input signal. This peak detector can be used to generate a reference voltage that falls within the middle of the input data, Fig. 18.28. The peak and valley detectors are used to find the minimum and maximum of the input signal. The two resistors average the minimum voltages and feed the result (the DC average of the input) to the bottom buffer circuit. The resistors also are used to leak charge off of the capacitors so that the averaging circuit can follow changes in the input data (the averaging is actually a *running average*). Figure 18.29 shows some simulation results. In the top plots the output signal from Fig. 18.26a is applied to the input of the DC generation circuit. As expected, the output of the DC generation circuit, after a start-up delay, goes right to the middle of the input data.

The speed (response time) of this generation circuit can be increased by reducing the resistors and capacitors in the circuit. However, if the values are reduced too much, a long string of zeroes or ones will cause the DC generation circuit's output (labeled "average" in Fig. 18.28) to go to ground or VDD . The bottom traces in Fig. 18.29 show the results of applying the output signal from Fig. 18.26b to the DC generation circuit. Again, the output moves quickly to the center of the data. What's more interesting in these figures is a comparison between the original input data and the regenerated output data (labeled *In* and *Out* in Fig. 18.29). The output data pulse widths are considerably different from the input pulse widths (remembering that there is a delay through both the channel RC seen in Fig. 18.26 and the buffer in Fig. 18.28). Ideally, the pulse widths (the data) of the input and output are the same. Looking at the responses in Fig. 18.26a, we see that the effect of the finite channel bandwidth is to distort the channel's output data. *The only solution to this problem is to make the channel transmission bandwidth effectively wider.* An equalizer (discussed in the next chapter) can be used for this purpose at the cost of reduced signal swing. We can also reduce the input impedance of the input buffer to lower the time constant associated with the channel. Let's discuss this second approach.

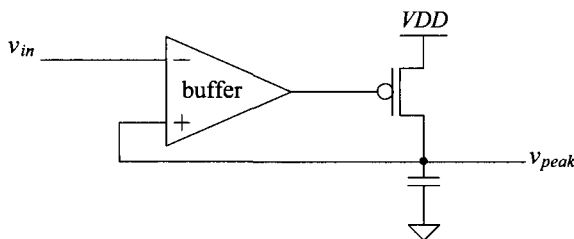


Figure 18.27 CMOS peak detector.

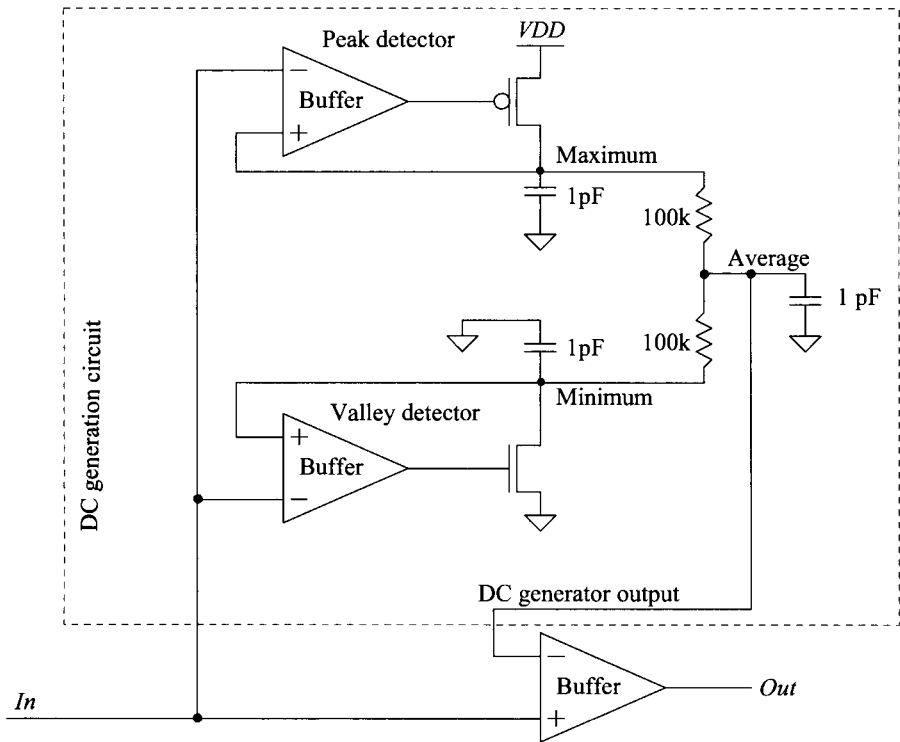


Figure 18.28 A DC generation circuit.

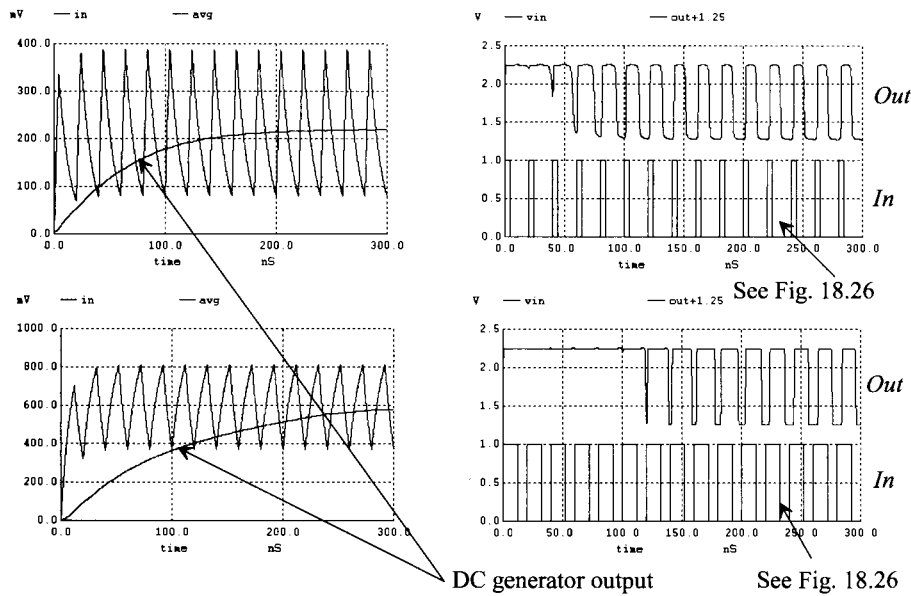


Figure 18.29 The output of the DC restore circuit in Fig. 18.28.

18.3.4 Reducing Buffer Input Resistance

A simple solution to increasing the effective bandwidth of the transmission channel is to reduce the input resistance of the input buffer. The time constant associated with the channel in Fig. 18.30 is 10 ns. When we connect the buffer to the transmission channel, the time constant drops to 3.33 ns (the three 1k resistors in parallel multiplied by the 10 pF capacitance). The drawbacks of this approach are increased power dissipation and reduced input signal amplitudes. Figure 18.31 shows how the buffer in Fig. 18.30 behaves with the input signals seen in Fig. 18.26a. The distortion (difference in the pulse widths of ones and zeroes when comparing the communication channel input, *In* , and the buffer output, *Out*) is much better.

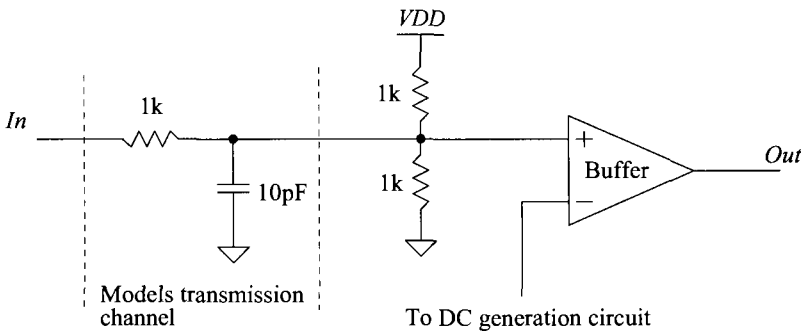


Figure 18.30 Reducing the input resistance of the input buffer.

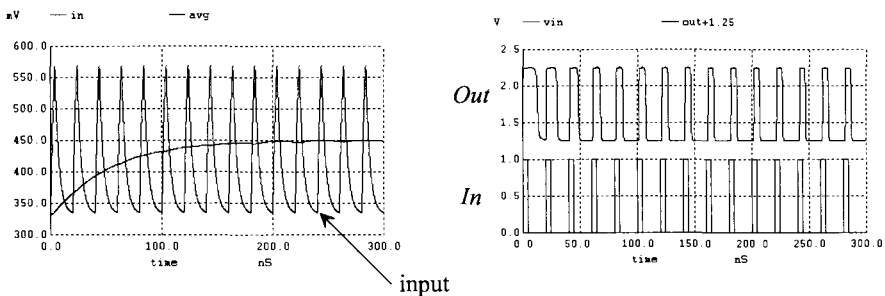


Figure 18.31 How reducing the input resistance of the buffer reduces the input signal amplitude (bad) and distortion (good).

Figure 18.32 shows an alternative low input resistance input buffer. When the input goes high, M1 turns on and M4 shuts off. This causes M2 to turn on and M5 to shut off. The output then goes high. When the input goes low, M1 turns off and M4 turns on (with M2 shutting off and M5 turning on). The result: the output goes low. Again, the power burned by M1 and M4 can be high if the transistors aren't sized properly. Further, the input swing is limited to a threshold voltage away from the supply voltages. This reduces the input voltage swing and further enhances the speed.

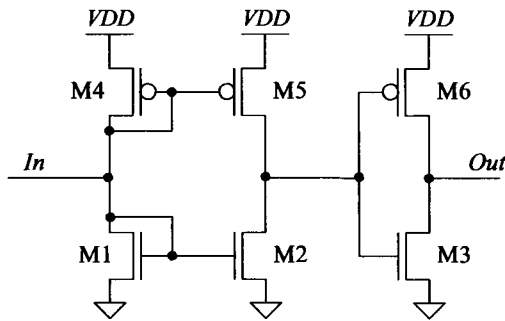


Figure 18.32 A low-input resistance input buffer.

18.4 Charge Pumps (Voltage Generators)

Often, when designing CMOS circuits, positive and negative DC voltages are needed that do not lie between ground and V_{DD} . An example of an application where a larger DC voltage source is needed was seen in Ch. 16 when we discussed Flash memory (see Fig. 16.59). A simple circuit, sometimes called a voltage pump (or more often, a *charge pump*), useful in generating a voltage greater than V_{DD} , is seen in Fig. 18.33.

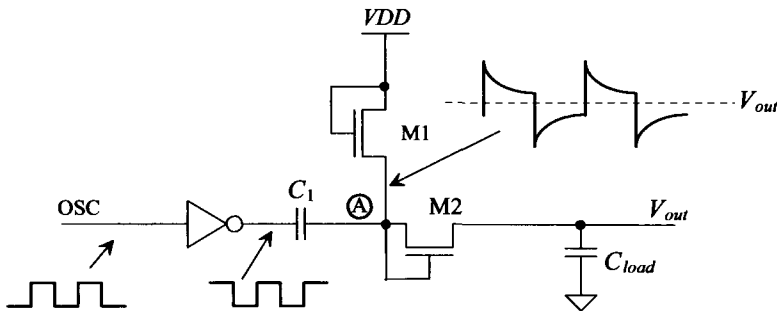


Figure 18.33 Pump used to generate a voltage greater than V_{DD} .

The operation of the voltage pump of Fig. 18.33 can be explained by first realizing that both M1 and M2 operate like a diode. M1 is simply used to pull point A to a voltage of $V_{DD} - V_{THN}$. M2 allows the charge from C_1 to charge C_{load} but not vice-versa. Let's begin the description of the circuit operation by assuming that the output of the inverter is low and point A is at a potential of $V_{DD} - V_{THN}$. When the output of the inverter goes high, the potential at point A increases to $V_{DD} + (V_{DD} - V_{THN}) = 2 \cdot V_{DD} - V_{THN}$. This turns on M2 and charges C_{load} to $2 \cdot (V_{DD} - V_{THN})$, [an extra V_{THN} because of M2's gate-source voltage drop], provided $C_1 \gg C_{load}$ and the oscillator frequency allows the capacitors to fully charge or discharge before changing states. In most practical situations, C_{load} and C_1 are comparable in size, and the output of the pump is loaded with a

DC load. The result is an output voltage with a startup time; that is, V_{out} does not immediately rise to $2 \cdot (V_{DD} - V_{THN})$ but requires several oscillator cycles to reach steady state. Also, the output has a ripple dependent on the DC load. Figure 18.34 shows the simulation results for the circuit of Fig. 18.13 using 10/1 NMOS (scale factor of 50 nm) with $C_{load} = C_1 = 1$ pF and an oscillator frequency of 10 MHz. Using this simple pump with a DC load, such as a resistor, may require employing larger MOSFETs and capacitors to avoid excessive voltage droop.

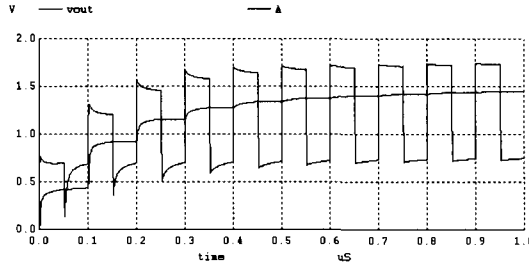


Figure 18.34 Simulating the operation of the voltage pump in Fig. 18.33.

Negative Voltages

The positive voltage pump uses n-channel MOSFETs, while the negative voltage pump, Fig. 18.35, uses p-channel MOSFETs. The reason for this comes from the requirement that the diode formed with the n+ (p+) implant used in the drain/source of the MOSFET combined with the p-substrate (n-well) does not become forward-biased. Forward biasing this parasitic diode is an *important concern* and is often the reason why some more exotic pump topologies can't be implemented. For example, in Fig. 18.35, we connect the well of the PMOS devices to ground instead of to their respective sources. Consider what would have happened had we connected the well to the source of the MOSFET. When the output voltage goes negative, the (n-type) well goes negative too. If the substrate (p-type) is at ground, this forward biases the n-well to substrate diode acting to clamp the output of the pump at a negative diode drop. While we could have left the bodies of the PMOS (the n-wells) tied to V_{DD} , here we chose to connect them to ground so that the body effect wasn't so severe (a lower threshold voltage).

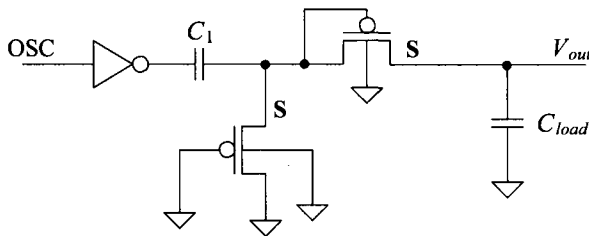


Figure 18.35 Negative voltage pump.

Using MOSFETs for the Capacitors

The capacitors used in Figs. 18.33 and 18.35 can be replaced with MOSFETs. An example of using n-channel MOSFETs in place of the capacitors in Fig. 18.33 is shown in Fig. 18.36. The main requirement on a MOSFET used as a capacitor is that its V_{GS} remain greater than V_{THN} at all possible operating conditions. In other words, the MOSFET must remain in the strong inversion region so that its capacitance is a constant $C'_{ox} \cdot W \cdot L$. The capacitor, C_{load} , in Fig. 18.36 remains in strong inversion because $V_{out} \gg V_{THN}$. The capacitor C_1 remains in strong inversion because, when the inverter output is low, the voltage on the gate of C_1 is $VDD - V_{THN}$ ($= V_{GS}$). When the output of the inverter is high (VDD), the voltage on the gate of C_1 is $2 \cdot VDD - V_{THN}$. In both cases (the output of the inverter high and low), $V_{GS} = VDD - V_{THN}$, and the MOSFET is in the strong inversion region.

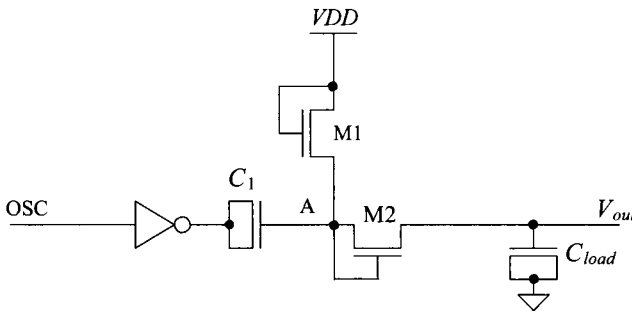


Figure 18.36 Using MOSFETs as capacitors.

18.4.1 Increasing the Output Voltage

Figure 18.37 shows a voltage pump with a higher output voltage. The increase in the output voltage comes from eliminating the threshold voltage drop at the gate and drain of M7. This allows the output to swing up to $2 \cdot VDD - V_{THN}$.

To understand the operation of the circuit, let's assume that the voltage at point A is low and the voltage at point C is $VDD - V_{THN}$. When the output of INV1 goes high, point A is VDD and the voltage at point C swings up to $2 \cdot VDD - V_{THN}$. This causes M4, M5, and M6 to turn on and pull points D and E to VDD . Now when point B goes high (point A goes back to zero), points D and E swing up to $2 \cdot VDD$ and the output goes to $2 \cdot VDD - V_{THN}$. Note that MOSFETs M2 and M3 are not needed; unless the pump drives a DC load, they never turn on. Also, separating points D and E is unnecessary unless the pump supplies a DC current.

18.4.2 Generating Higher Voltages: The Dickson Charge Pump

The voltage pumps of the last section are limited to voltages less than $2 \cdot VDD$ and greater than $-2 \cdot VDD$. Figure 18.38 shows a scheme for generating arbitrarily high voltages (limited by the breakdown voltage of the capacitors or the oxide breakdown of the MOSFETs). Again, the MOSFETs are used as diodes in this configuration.